

Отчет подобия

📁 GQW (Graduation qualification work)

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Mustafo Aminjonov

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Тревога

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Замена букв	ⓑ	1
Интервалы	A→	0
Микропробелы	␣	0
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Источники

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“Systems analysis and mathematical modeling” department
Aminjonov Mustafo Anvarjon ugli
Theme: “The impact of Digitalization and the Adoption of High Technologies on Economic Growth in Central Asian Countries (Evidence from Kazakhstan, Uzbekistan, and Kyrgyzstan)”

60412000- “International economics and management” (by regions and lines of activity)”
For a bachelor's degree in education
Graduate thesis

The work was reviewed and recommended for defense:

Scientific supervisor:
Candidate of physical and mathematical sciences, associate professor
Akabirkhodjayeva D.R.
_____ (signature)
Head of the department:
Doctor of physical and mathematical sciences, professor
Raimova G.M.
_____ (signature)
“__” _____ 2026 year.
Tashkent-2026

Annotatsiya

Mazkur bitiruv malakaviy ishida raqamlashtirish va yuqori texnologiyalarni joriy etishning Markaziy Osiyo mamlakatlari iqtisodiy o’sishiga ta’siri Qozog’iston, O’zbekiston va Qirg’iziston misolida tadqiq etilgan. Ishda raqamlashtirish jarayonlari faqat texnologik yangilanish sifatida emas, balki iqtisodiy modernizatsiya, mahsuldorlik, innovatsion faollik va raqobatbardoshlikni oshirishga xizmat qiluvchi strategik omil sifatida ko’rib chiqilgan. Tadqiqotda iqtisodiy o’sish va raqamlashtirishga oid ilmiy-nazariy yondashuvlar o’rganilgan, uch mamlakatning raqamli rivojlanish ko’rsatkichlari qiyosiy tahlil qilingan hamda Qozog’istonning mintaqaviy benchmark sifatidagi tajribasi asoslab berilgan.

Empirik qismda 2004-2023 yillar bo’yicha panel ma’lumotlar asosida raqamlashtirish indikatorlari va iqtisodiy o’sish o’rtasidagi bog’liqlik baholangan. Tadqiqotda internetdan foydalanuvchilar ulushi, mobil aloqa obunalari, keng polosali internet, savdo ochiqligi va inflyatsiya kabi ko’rsatkichlardan foydalanilgan. Ekonometrik tahlil natijalari raqamlashtirishning iqtisodiy o’sishga ta’siri ko’proq uzoq muddatli, bilvosita va tarkibiy omillar orqali namoyon bo’lishini ko’rsatadi. Ish yakunida O’zbekiston uchun raqamli infratuzilmani rivojlantirish, kichik va o’rta biznesda raqamli texnologiyalarni keng joriy etish, inson kapitali va raqamli ko’nikmalarni kuchaytirish bo’yicha amaliy tavsiyalar ishlab chiqilgan.

Kalit so’zlar: raqamlashtirish, yuqori texnologiyalar, iqtisodiy o’sish, AKT, panel ma’lumotlar, Qozog’iston, O’zbekiston, Qirg’iziston, raqamli infratuzilma, innovatsiya.

Аннотация

В данной выпускной квалификационной работе исследуется влияние цифровизации и внедрения высоких технологий на экономический рост стран Центральной Азии на примере Казахстана, Узбекистана и Кыргызстана. В работе цифровизация рассматривается не только как процесс технологического обновления, но и как важный фактор экономической модернизации, повышения производительности, развития инновационного потенциала и укрепления национальной конкурентоспособности. Особое внимание уделено теоретическим подходам к взаимосвязи технологий и экономического роста, сравнительному анализу цифрового развития выбранных стран, а также обоснованию роли Казахстана как регионального benchmark-примера. В эмпирической части исследования использованы панельные данные за период 2004-2023 годов. Для оценки связи между цифровизацией и экономическим ростом были рассмотрены такие показатели, как доля пользователей интернета, количество мобильных подключений, фиксированный широкополосный интернет, открытость торговли и инфляция.

Эконометрический анализ показал, что воздействие цифровизации на экономический рост не всегда проявляется непосредственно в краткосрочном периоде, поскольку оно во многом зависит от качества инфраструктуры, уровня человеческого капитала, деловой среды и способности экономики использовать цифровые технологии продуктивно. На основе результатов исследования разработаны практические рекомендации для Узбекистана, направленные на развитие цифровой инфраструктуры, расширение цифровизации малого и среднего бизнеса, укрепление взаимодействия между университетами, IT Park и частным сектором, а также сокращение регионального цифрового разрыва.

Ключевые слова: цифровизация, высокие технологии, экономический рост, ИКТ, панельные данные, Казахстан, Узбекистан, Кыргызстан, цифровая инфраструктура, инновации.

Abstract

This graduation qualification work examines the impact of digitalization and the adoption of high technologies on economic growth in Central Asian countries, with a focus on Kazakhstan, Uzbekistan and Kyrgyzstan. The study treats digitalization not merely as a technological process, but as a strategic driver of economic modernization, productivity improvement, innovation capacity and long-term competitiveness. The theoretical part of the research reviews key approaches linking technological progress and economic growth, while the comparative part evaluates the digital transformation paths of the selected countries and explains Kazakhstan’s role as a regional benchmark.

The empirical analysis is based on panel data covering the period from 2004 to 2023. The study uses indicators such as internet users, mobile cellular subscriptions, fixed broadband subscriptions, trade openness and inflation to assess the relationship between digitalization and economic growth. The econometric results suggest that the effect of digitalization on economic growth may not always appear as a direct short-term impact. Instead, it is more likely to operate through long-term and indirect channels, including infrastructure quality, human capital, innovation, business digital adoption and institutional capacity. Based on the findings, the thesis develops practical recommendations for Uzbekistan, including improving digital infrastructure, supporting digital adoption among small and medium-sized enterprises, strengthening cooperation between universities, IT Park and the private sector, reducing regional digital inequality, and developing digital skills.

Keywords: digitalization, high technologies, economic growth, ICT, panel data, Kazakhstan, Uzbekistan, Kyrgyzstan, digital infrastructure, innovation.

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INTRODUCTION

Relevance of graduation qualification work. In recent years, digitalization and the usage of cutting-edge technologies have become main players of economic development and strategic change in the whole world. According to data from the World Bank and IMF, the global share of people using the internet jumped from 16% in 2005 to 67% in 2022, which means that digital connectivity demonstrated the rapid growth. During the same period, countries with high levels of ICT (Information and communication technology) sensitivity have overall showed powerful productiveness in terms of improvement and also greater integration in global economy. For countries that are located in Central Asia, digital growth has improved differently according to statistics and this is the problem which is relevant.

If we look at the data from World Bank, the internet usage of Kazakhstan stands for 91% in 2022, which is a little higher compared to Uzbekistan with 77% and 58% in Kyrgyzstan. This big difference in digital penetration identifies differences in entrance to digital technologies and recommends potential diverse impacts on economic growth inside the region. If we look from economic viewpoint, Central Asian economies still have problems with productivity and diversification. World Bank suggests that average GDP per capita rise rates in the region have been unstable over the last 20 years, copying sensitivity to the external shocks and limited technological promotion. In contrast, investment in digital infrastructure has risen at the same time by the growth in mobile subscriptions going to 100 subscribers per 100 people in Kazakhstan by the late 2010s, compared to Uzbekistan and Kyrgyzstan which is lower but have steadily growing levels. These trends points to digital technologies are increasingly fixed in terms of economy, however, their macroeconomic impact stays measured inadequately.

There is clear misunderstanding between the size of digital investments and the limited access of empirical information emphasizing their benefits to economic growth in Central Asia, despite the developed policy on digital transformation. International studies show that there is positive relationship between economic growth and ICT development in advanced and developing economies, while the region stays unchanged in cross-country econometric analyses. Startegic development and digital transformation priorities of Central Asian economies, carrying this research nowadays is particularly vital. An econometric analysis covers the period between 2004 and 2023 based on official statistics, which helps to get result more accurate of the impact of digitalization on economic growth.

Degree of study of the research problem. In modern economic literature, the connection between digitalization, information and communication technology (ICT), and economic growth has been widely studied from both theoretical and empirical angles. One of the foundational theoretical contributions to understanding technology-driven growth is associated with Paul Romer, whose endogenous growth theory emphasizes the role of technological progress, innovation, and knowledge spillovers as internal drivers of long-term economic growth. Philippe Aghion and Peter Howitt provided additional theoretical advancements by introducing Schumpeterian growth models that emphasized the significance of technological competition and creative destruction. Since ICT adoption frequently results changes in structure, productivity shifts, and the development of new economic sectors, these models are especially essential for studying digital technologies.

From an information economics perspective, the works of Hal Varian and Carl Shapiro made a great contribution to understanding digital marketplaces, network effects, and the financial consequences of information technology. Their study provides crucial conceptual tools for examining the digital economy by elucidating how digital platforms, connectivity, and information asymmetries affect market efficiency and economic performance.

Panel regression models, comparative analysis and fixed effects estimations included in methodological approaches help researchers to control for country-specific characteristics and identify the small effect of digitalization on economic expansion. Building on these well-established scientific methods, this work uses widely known indicators to apply panel econometric methods to annual data for Kazakhstan, Uzbekistan, and Kyrgyzstan from 2004 to 2023.

The object of the graduation qualification work. The research object of this graduation work is the process of economic growth in Central Asian countries, including Kazakhstan, Uzbekistan and Kyrgyzstan within the context of digital transformation and technological development.

The subject of graduation qualification work. The research subject of this work includes the impact of information and communication technology (ICT) and digitalization in particular usage and mobile connectivity on economic growth. Moreover, the economic relationships and mechanisms via which digital technologies affect growth dynamics in the choosen countries.

Aim of graduation qualification work. The aim of this work is to study how digitalization and information and communication technologies affect on economic growth in Central Asian countries at the same time define the major factors impacting growth dynamics and also support scientific recommendations for strengthening Uzbekistan’s digital development strategy using econometric analysis.

Tasks of graduation qualification work. In order to achieve the goal, following steps are set:

1. to explore scientific views and theoretical structures related to digitalization, ICT and economic growth;
2. to examine the current state and development trends of digitalization in Central Asian countries;
3. to do an a statistical econometric analysis to asses how digitalization has impacted economic growth between 2004 and 2023;
4. to use the economic data to identify important technological problems affecting the growth of the economy;

5. to apply international knowledge to develop and evaluate practical suggestions for improving Uzbekistan's digital development policy;
6. to collect and organize statistics on economic development measures and digitalization for Kazakhstan, Uzbekistan and Kyrgyzstan.

Methodology of graduation qualification work. A detailed methodological strategy which combines empirical, statistical and theoretical study methods are used in this work. Economic growth and scientific ideas on ICT and digitalization development are analyzed at the theoretical step through synthesis, generalization analysis. The study's scientific element relies on studies and comparisons of macroeconomic and digital indicators for Kazakhstan, Uzbekistan, and Kyrgyzstan from 2004 to 2023. Here is used panel data econometric modeling for statistical and mathematical methods in order to assess how digitalization impacts on economic growth. In order to find important correlations between variables regression analysis is implemented with pooled OLS and fixed effects models. Another thing is, for evaluation empirical findings and develop beneficial recommendations for enhancing Uzbekistan's digital development plan, methods of modeling are used. All the statistics for the analysis are taken from international databases, such as the World Bank and also reports from the IMF (International monetary fund) and ITU (International telecommunication union).

The structure of graduation qualification work. Graduation qualification work consists of 3 chapters, they are following:

Chapter I focuses to theoretical and methodological foundations of digitalization and economic growth. The key concepts of digitalization, information and communication technologies and economic growth are studied too in this chapter.

Chapter II mainly focuses on international experience and empirical analysis. This chapter shows the reason for choosing Kazakhstan as a benchmark country and gives comparative analysis of digitalization numbers in Central Asia countries. Moreover, chapter II demonstrates econometric analysis based on panel data for Kazakhstan, Uzbekistan and Kyrgyzstan over the 2004-2023.

Chapter III is focused on adaptation of international experience and strategic advises for Uzbekistan. The current condition of digitalization in Uzbekistan is also examined in this chapter with international experience. Also practical recommendations are given to improve the role of digitalization in helping economic growth.

CHAPTER I. METHODOLOGICAL AND THEORETICAL FOUNDATIONS OF ECONOMIC GROWTH AND DIGITALIZATION

- Definitions of ICT, digitalization and high-technology adoption

In the era of globalization and technological advancement, digital technologies have become integral to the socio-economic fabric of nations worldwide. The proliferation of the internet, mobile technology, and digital communications has reshaped the dynamics of economies, influencing growth patterns and development strategies. This transformation is particularly notable in regions with diverse economic backgrounds, such as Central Asia and Europe, where digital technology penetration varies significantly. Understanding the impact of digital technologies on economic growth is essential for policymakers and stakeholders. While extensive research exists on the global impact of digitalization, there is a need for focused studies that examine specific regions with diverse economic landscapes. Central Asia and Europe present an interesting case due to their varied levels of economic development, digital infrastructure, and policy environments.

Information and Communication Technologies (ICT) constitute one of the most transformative technological forces in contemporary economic development. In broad academic terms, ICT refers to a complex system of technologies used for the collection, storage, processing, transmission, and dissemination of information through electronic means. These technologies include telecommunications networks, internet infrastructure, computing systems, software platforms, data-processing systems, and digital communication tools.

Information and Communication Technologies (ICT) represent a broad set of technologies used for the collection, storage, processing and transmission of information through electronic systems. In economic literature, ICT is commonly understood as a combination of telecommunications infrastructure, computing systems, software, and digital networks that enable economic coordination and value creation. According to the World Bank, ICT plays a fundamental role in reducing transaction costs, increasing productivity, and improving market access in both developed and developing economies[1].

Information and Communication Technologies (ICT) constitute one of the most transformative technological forces in contemporary economic development. In broad academic terms, ICT refers to a complex system of technologies used for the collection, storage, processing, transmission, and dissemination of information through electronic means. These technologies include telecommunications networks, internet infrastructure, computing systems, software platforms, data-processing systems, and digital communication tools.

The Information and Communication Technology (ICT) sector is the backbone of digital transformation. In addition, while the digitalisation of the economy extends to all sectors to varying degrees, it is particularly prevalent in the Content and Media sector. These two highly digital sectors together comprise the Information Industries, a key aggregate that has been widely used to assess digital performance across countries. The growing significance of these industries underscores the need for well-designed definitions to enable cross-country comparable measurement, which in turn supports evidence-based digital policymaking[2].

The history of ICT

The economic importance of ICT began to attract serious academic attention in the late twentieth century, particularly after the productivity paradox debate initiated by Robert Solow. In his seminal work on technological change and economic growth, Solow (1957) emphasized that long-term output growth cannot be explained solely by capital and labor accumulation; instead, technological progress plays a decisive role. From a theoretical perspective, ICT is frequently analyzed within the framework of technological progress and productivity growth. Robert Solow emphasized the importance of technological progress in long-term economic growth, arguing that increases in output cannot be explained solely by capital and labor accumulation. Later empirical research extended this idea by identifying ICT as a key driver of Total Factor Productivity [3](TFP). By the early 2000s, empirical studies began systematically measuring ICT capital and its contribution to Total Factor Productivity (TFP). Jorgenson and Stiroh (2000) demonstrated that investment in ICT capital significantly contributed to productivity growth in the United States during the 1990s. Their research provided quantitative evidence that ICT investments were not merely complementary assets but core drivers of economic expansion[4].

Figure 1.1.1. Individuals using the Internet (%)

Source: HYPERLINK "<https://www.itu.int/en/ITU-D/Statistics/Pages/stat/default.aspx>" <https://www.itu.int/en/ITU-D/Statistics/Pages/stat/default.aspx>

ITU estimates that approximately 6 billion people - or 74 per cent of the world's population - are using the Internet in 2025. This represents an increase from 60 per cent in 2020, with 1.3 billion people estimated to have come online during that period. However, this leaves 2.2 billion people still offline.

While ICT refers to technological tools and infrastructure, the concept of digitalization reflects a broader systemic transformation. Digitalization involves the integration of digital technologies into economic, organizational, and social processes, fundamentally altering how value is created and delivered. The distinction between digitization and digitalization is important. Digitization refers to converting analog information into digital form. Digitalization, however, implies structural change - transformation of business models, production systems, governance mechanisms, and labor organization.

Digitalization

The concept of digitalization goes beyond simple digitization. While digitization refers to converting information into digital form, digitalization describes the integration of digital technologies into economic and organizational processes. Erik Brynjolfsson and Andrew McAfee argue that digitalization transforms business models, labor markets, and productivity patterns by enabling automation, data analytics, and network-based coordination. Their work highlights how digital technologies reshape entire industries rather than merely improving existing processes[5].

The economics of digitalization cannot be fully understood without examining network effects and platform markets. Varian and Shapiro (1999), in Information Rules, demonstrate how digital goods exhibit increasing returns to scale and strong network externalities[6]. Digital platforms create value by connecting users, reducing search costs, and facilitating transactions. Network effects imply that the value of a digital service increases as more users adopt it. This characteristic accelerates digital diffusion and market concentration.

Brynjolfsson and McAfee (2014), in The Second Machine Age, argue that digital technologies have reached a stage where they fundamentally reshape productivity patterns, enabling automation, artificial intelligence, and data-driven decision-making. According to their analysis, digital transformation differs from previous technological revolutions because of its exponential development speed and scalability[7].

The OECD defines the digital economy as an economy that is significantly enhanced by digital technologies, data flows, and digital platforms[8]. The digital economy is characterized by:

- 1. High reliance on data as a production factor
- 2. Platform-based market structures
- 3. Network effects and scalability
- 4. Rapid diffusion of innovation

In the context of Central Asia, digitalization is closely connected with expanding internet penetration, mobile connectivity, and integration into global digital markets. Therefore, digitalization is both a technological and institutional process requiring supportive infrastructure, human capital, and regulatory frameworks.

Country Year Individuals using the internet (% of population)

Kyrgyz Republic	2004	5.1
2005	10.5	
2006	12.3	
2007	14	
2008	15.7	
2009	16	
2010	16.3	
2011	17.5	
2012	19.8	
2013	23	
2014	28.3	
2015	30.2	
2016	40.5	
2017	50.8	
2018	62.8	
2019	64.1	
2020	72	
2021	77.1	
2022	82.6	
2023	88.5	
Kazakhstan	2004	2.7
2005	3	
2006	3.3	
2007	4	
2008	11	
2009	18.2	
2010	31.6	
2011	50.6	
2012	61.9	
2013	63.3	
2014	66	
2015	70.8	
2016	74.6	
2017	76.4	
2018	78.9	
2019	81.9	
2020	85.9	
2021	90.9	
2022	92.3	
2023	92.9	
Uzbekistan	2004	2.6
2005	3.3	
2006	6.4	
2007	7.5	

2008	9.1
2009	11.9
2010	15.9
2011	18.6
2012	23.6
2013	26.8
2014	35.5
2015	42.8
2016	46.8
2017	48.7
2018	55.2
2019	70.4
2020	71.1
2021	76.6
2022	83.9
2023	89

Table 1.2.1. Individuals using the internet (% of population)
Source:<https://databank.worldbank.org/reports.aspx?source=2&type=metadata&series=IT.NET.USER.ZS>

Understanding of high-technology

High technology (high-tech) is generally defined based on technological intensity and R&D investment. The OECD classification system categorizes industries according to R&D expenditure relative to value added, distinguishing high-technology, medium-high, medium-low, and low-technology sectors[9].

High-technology industries typically include:

1. Advanced telecommunications
2. Computer and electronic manufacturing
3. Biotechnology
4. Aerospace
5. Artificial intelligence and advanced software

Romer (1990) revolutionized growth theory by introducing endogenous technological change, arguing that knowledge accumulation and innovation generate increasing returns and sustained growth. Unlike neoclassical models, endogenous growth theory treats technology as an internal outcome of economic decisions, particularly investment in human capital and R&D. The notion of high technology is commonly defined using technology-intensity classifications. The OECD classifies industries as high-technology based on their R&D intensity and knowledge content. In growth economics, high-technology adoption refers to the diffusion and effective utilization of advanced technologies such as broadband infrastructure, artificial intelligence, cloud computing, and digital platforms. Paul Romer argues in endogenous growth theory that technological diffusion and knowledge spillovers generate increasing returns and sustained economic growth[10].

Figure 1.3.1.
Conceptual framework linking digitalization and economic growth.

The theoretical relationship between economic growth and digitization can be seen in the theoretical structure shown in Figure 1.1.1. The basis of technical development is digitalization, which is seen in the growth of ICT infrastructure, including internet access and mobile connectivity. Neoclassical and endogenous growth ideas suggest that developments in technology increase productivity through minimizing transaction costs, increasing efficiency, and aiding the spread of knowledge (Solow, 1957; Romer, 1990). According to the theory, digital infrastructure supports innovation and increases productivity by simplifying the adoption of technological advances, information access, and integration into global supply chain by both companies and individuals.

According to the Schumpeterian growth model (Aghion & Howitt, 1992), innovation is used as an indicator for technology adoption, which results in structural transformation and long-term economic growth. Moreover, digitization increases trade integration and accumulation of capital by removing market barriers and increasing connectivity, thus boosting investment and global competitiveness. Finally, all of these factors lead to higher GDP per capita growth, which is used as primary indicator of economic performance.

1.2. The role of technology and economic growth theories

Understanding the role of technology in economic growth has been one of the central questions in economic theory. From classical economics to modern endogenous growth models, technological progress has been identified as a key driver of productivity improvements, structural transformation, and long-term economic expansion. In the context of digitalization and high-technology adoption, it is necessary to examine how economic growth theories conceptualize technological change and how these theoretical foundations justify the empirical analysis conducted in this study.

Neoclassical Growth Theory and Exogenous Technological Progress

The neoclassical growth model, primarily developed by Robert Solow (1956; 1957), laid the foundation for modern growth theory. Solow introduced a production function in which output depends on capital, labor, and technology. In his framework, technological progress is treated as an exogenous factor that shifts the production function upward over time[11]. According to Solow (1957), increases in capital accumulation alone cannot explain long-term economic growth. Instead, the residual component-commonly referred to as "total factor productivity" (TFP)-captures the contribution of technological progress[12].

Empirical studies based on the Solow model demonstrated that a substantial portion of economic growth in developed economies was attributable to technological improvements rather than merely increases in labor or capital inputs. In the context of digitalization, ICT infrastructure, internet penetration, and mobile connectivity can be interpreted as components of technological progress that enhance productivity. As digital technologies reduce transaction costs, improve coordination, and increase information accessibility, they contribute to higher output levels. Thus, from a neoclassical perspective, digitalization acts as an exogenous productivity-enhancing factor influencing GDP growth. However, a major limitation of the Solow model is

that it does not explain how technological progress is generated. Technology is assumed to grow independently of economic decisions. This limitation led to the development of endogenous growth models.

Endogenous Growth Theory and Innovation Dynamics

Endogenous growth theory emerged in the late 1980s and early 1990s to address the shortcomings of exogenous models. Paul Romer (1990) introduced a framework in which technological change results from intentional investment in research and development (R&D) and human capital accumulation[13]. In Romer’s model, knowledge spillovers generate increasing returns to scale, allowing economies to sustain long-term growth. Unlike the Solow model, where technology is external, endogenous models treat innovation as an internal outcome of economic incentives. Firms invest in R&D because knowledge creation leads to new products and productivity gains. In digital economies, investments in ICT infrastructure and digital platforms stimulate innovation ecosystems and generate spillover effects across industries.

Aghion and Howitt (1992) further developed the Schumpeterian growth model, emphasizing “creative destruction” as the core mechanism of technological progress[14]. According to this approach, economic growth results from continuous innovation cycles, where new technologies replace outdated ones. Digital transformation represents a modern form of creative destruction, as traditional production processes are replaced by automated, data-driven systems. Empirical research supports the relevance of endogenous growth mechanisms in digital economies. Studies show that ICT adoption increases firm productivity, enhances innovation capacity, and fosters knowledge diffusion[15]. For developing economies, digital infrastructure reduces barriers to entry and facilitates integration into global markets. Therefore, endogenous growth theory provides a strong theoretical foundation for examining how digitalization indicators influence GDP per capita growth in Central Asian countries.

ICT as a General-Purpose Technology

Another important theoretical perspective is the concept of General-Purpose Technologies (GPTs). Bresnahan and Trajtenberg (1995) define GPTs as technologies that have broad applicability across sectors and generate complementary innovations[16]. Historically, examples include electricity and the steam engine. In the modern era, ICT is widely recognized as a general-purpose technology. ICT affects multiple sectors simultaneously by improving communication efficiency, enabling automation, and facilitating digital platforms. According to the OECD (2023), digital technologies enhance productivity by enabling firms to optimize supply chains, adopt cloud computing solutions, and leverage big data analytics[17].

In emerging economies such as Kazakhstan, Uzbekistan, and Kyrgyzstan, digitalization may accelerate convergence with advanced economies by reducing information asymmetries and lowering transaction costs. This aligns with empirical findings suggesting that ICT infrastructure positively correlates with economic growth, particularly when complemented by institutional quality and human capital development[18].

Digitalization, Productivity, and Structural Transformation

Beyond aggregate growth models, digitalization influences microeconomic efficiency and sectoral dynamics. According to the World Bank (2016), digital technologies reduce coordination costs, increase market transparency, and foster competition[19]. These mechanisms enhance allocative efficiency and resource utilization. Brynjolfsson and McAfee (2014) argue that digital technologies represent a “second machine age,” characterized by automation, artificial intelligence, and data-driven decision-making[20]. These technologies increase labor productivity but may also alter employment structures. Consequently, digital transformation contributes to structural change by shifting labor from low-productivity sectors to high-value-added industries.

From a macroeconomic perspective, digitalization enhances total factor productivity (TFP), which remains the main long-term driver of economic growth. Empirical analyses conducted by the IMF (2018) demonstrate that countries with higher ICT penetration tend to experience stronger productivity growth, particularly when supported by favorable regulatory environments[21]. Thus, digitalization affects growth through multiple channels: productivity enhancement, innovation stimulation, trade integration, and capital accumulation.

Numerous empirical studies confirm the positive relationship between digitalization and economic growth. For instance, Czernich et al. (2011) found that broadband infrastructure significantly increases GDP growth rates in OECD countries[22]. Similarly, Qiang, Rossotto, and Kimura (2009) demonstrated that a 10 percent increase in broadband penetration can raise GDP growth by up to 1.38 percentage points in developing economies[23]. In emerging markets, the growth effect of digitalization depends on complementary factors such as institutional quality, education, and trade openness. This is particularly relevant for Central Asia, where structural reforms and digital policies are evolving.

1.3. Research Methodology and Analytical Approaches to Estimating the Impact of Digitalization on Economic Growth

The empirical assessment of the relationship between digitalization and economic growth requires a structured methodological approach grounded in economic theory and quantitative analysis. Given the macroeconomic nature of this study and the availability of longitudinal data for Kazakhstan, Uzbekistan, and Kyrgyzstan over the period 2004-2023, the research design integrates descriptive statistics, comparative analysis, and panel econometric modeling.

Research Design and Data Structure

This work adopts a quantitative research design based on secondary macroeconomic data. The analysis covers three Central Asian countries- Kazakhstan, Uzbekistan, and Kyrgyzstan-over a 20-year period (2004-2023). The selection of this time frame allows for capturing structural digital transformation processes and ensuring sufficient observations for panel econometric estimation. The empirical approach follows the tradition of growth regressions widely applied in macroeconomic research[24]. Panel data methodology is particularly appropriate for this study because it combines cross-country and time-series dimensions, increasing estimation efficiency and controlling for unobserved heterogeneity[25].

Panel data models are commonly used in empirical growth literature when analyzing the impact of technological and institutional variables on economic performance[26]. Compared to pure cross-sectional or time-series models, panel regression allows controlling for country-specific characteristics that remain constant over time but may influence growth outcomes.

Data Sources and Variable Selection

The empirical analysis relies primarily on internationally recognized statistical databases to ensure reliability and comparability.

The main data sources include:

- 1. World Bank - World Development Indicators (WDI)
- 2. International Telecommunication Union (ITU)
- 3. International Monetary Fund (IMF)
- 4. Worldwide Governance Indicators (WGI)
- 5. The dependent variable in this study is:
- 6. GDP per capita growth (annual %)

This indicator reflects improvements in economic performance adjusted for population changes and is widely used in empirical growth literature[27].

The key explanatory variables representing digitalization include:

- 1. Internet users (% of population)
- 2. Mobile cellular subscriptions (per 100 people)

These indicators are standard proxies for ICT penetration and digital adoption[28]

Control variables are included to account for additional determinants of economic growth:

1. Trade openness (% of GDP)
2. Inflation (consumer prices, annual %)

The inclusion of control variables follows established growth regression practices to reduce omitted variable bias[29].

Econometric Model Specification

The empirical estimation is based on an augmented growth regression model derived from the Cobb-Douglas production function framework. In line with the Solow growth model and its empirical extensions[30].

Panel Data Estimation Techniques

Three main estimation approaches are considered:

1. Pooled Ordinary Least Squares (OLS)

The pooled OLS model treats all observations as homogeneous and ignores country-specific heterogeneity. While simple, this method may lead to biased estimates if unobserved country characteristics influence growth[31].

2. Fixed Effects Model (FE)

The fixed effects estimator controls for time-invariant country-specific factors, such as institutional structure or geographic characteristics[32]. Given the structural differences between Kazakhstan, Uzbekistan, and Kyrgyzstan, the fixed effects model is particularly suitable for this analysis.

3. Random Effects Model (RE)

The random effects model assumes that country-specific effects are uncorrelated with explanatory variables. The Hausman test is used to determine whether fixed or random effects provide consistent estimates[33].

In macroeconomic panel studies, fixed effects models are often preferred when analyzing a limited number of countries over time, as they provide more robust control for heterogeneity[34].

Analytical Justification for the Methodological Approach

The use of panel regression in this study is theoretically justified by endogenous growth theory, which suggests that technological adoption influences long-term productivity and growth. Empirical literature demonstrates that ICT penetration significantly contributes to GDP growth, particularly when combined with investment and trade integration[35]. Czernich et al. (2011) find that broadband expansion positively affects economic growth in OECD countries[36]. Similarly, Qiang et al. (2009) show that ICT infrastructure has stronger growth effects in developing economies when institutional conditions are supportive[37]. Therefore, applying panel econometric techniques to Central Asian countries allows testing whether similar mechanisms operate in transition and emerging economies.

CHAPTER II. INTERNATIONAL EXPERIENCE AND EMPIRICAL ANALYSIS OF DIGITALIZATION AND ECONOMIC GROWTH

2.1. Kazakhstan as a benchmark economy in Central Asia: digital transformation trajectory, economic outcomes, and policy lessons

Kazakhstan is treated as the benchmark

Kazakhstan is treated as the benchmark country among Kazakhstan, Uzbekistan, and Kyrgyzstan not because it is free from digital problems, but because it represents the most institutionally mature and strategically consistent case of digital transformation among the three. OECD’s 2023 assessment describes Kazakhstan as a country that has been building an ambitious digitalization agenda since 2013 and states that the country has already developed a comprehensive digital government system while beginning to establish the legal and regulatory conditions for the digital transition of the wider economy[38]. This is a stronger foundation than in the other two selected countries, where digital reforms are important but institutionally less consolidated. In addition, Kazakhstan remains the largest and highest-income economy among the three: according to World Bank country data for 2024, its GDP reached USD 291.48 billion and GDP per capita USD 14,154.6, compared with USD 114.97 billion and USD 3,161.7 in Uzbekistan, and USD 17.48 billion and USD 2,420.2 in the Kyrgyz Republic. From a methodological point of view, this makes Kazakhstan the most suitable reference case for evaluating how digitalization and high-technology adoption may interact with economic growth in Central Asia[39]. The idea of using Kazakhstan as a benchmark is also supported by the regional and academic literature. Marco Bianchini and Insung Kwon, in the OECD regional study Private Sector Digitalisation in Central Asia, treat Kazakhstan as the most advanced private-sector digitalization case in the region, while also showing that its transformation is still incomplete[40]. At the level of public governance research, Baurzhan Bokayev and co-authors analyze Kazakhstan’s e-government reforms through a citizen-centric lens and show that the country has already moved beyond a basic administrative digitalization stage toward broader service transformation. Gul Jussupova, Bokayev, and Dauletbay Zhussip likewise discuss Kazakhstan’s experience in terms of digital government maturity, emphasizing both progress and unresolved weaknesses in the public-sector digital model. In other words, Kazakhstan has already generated enough institutional experience and scholarly attention to function as a benchmark, while Uzbekistan and Kyrgyzstan remain more clearly in the phase of catch-up and selective adaptation[41].

A further reason for selecting Kazakhstan is that it is similar enough to Uzbekistan and Kyrgyzstan to allow comparison, but advanced enough to make benchmarking meaningful. All three countries are post-Soviet, landlocked, transition economies, and all three rely on state-led modernization to a significant extent. At the same time, Kazakhstan has had more time to institutionalize digital transformation, more fiscal and administrative capacity to scale reforms, and a stronger record in international rankings. This balance between comparability and differentiation is essential in a graduation thesis. If Kazakhstan were too different, it would not be a useful benchmark. If it were too similar, it would not show a meaningful gradient of digital maturity. Instead, Kazakhstan occupies the precise middle ground needed for benchmarking: close enough to be relevant, yet sufficiently ahead to illuminate the path of regional digital development[42].

Evolution of Kazakhstan’s digital strategy

Kazakhstan’s digital strategy did not begin with a single isolated programme. OECD notes that the country’s digital transformation of the economy “started a few years ago” and that in 2013 Kazakhstan was already among the first countries in the region to adopt a digital programme as part of economic policy. This matters because benchmark status is not created by a single project; it is created by policy continuity. The first phase of Kazakhstan’s digital trajectory was therefore characterized by the recognition that digitalization should be linked not only to public services, but also to broader economic modernization. This early start gave Kazakhstan an institutional advantage over the other two countries, whose more visible digital strategies came later and under somewhat narrower initial conditions. The second major phase was the Digital Kazakhstan Strategy, which OECD describes as having been launched in 2018 and as successful in developing a comprehensive digital government system and starting to establish the legal and regulatory conditions for the digital transition of the Kazakh economy.

The OECD regional paper further explains that the Digital Kazakhstan programme (2018-2022) treated the digitalization of the economy as one of its key components and focused especially on extractive industries and manufacturing, where technologies capable of raising productivity were prioritized.

The evolution of Kazakhstan's digital strategy also included supporting programmes and ecosystem-building tools. OECD notes the launch of the Astana Hub International Technology Park in 2018 and highlights the cybersecurity initiative Cybershield Kazakhstan, which later evolved further. In early 2022, according to the same OECD report, Kazakhstan launched Digital Era Lifestyle (DigitEL), described as its third five-year digitalization programme, with a focus on quality and safe internet, using ICT businesses as a growth lever, and further digitalizing the service sector. This continuity is important. It shows that Kazakhstan's benchmark role does not rely only on past achievements; it also relies on an evolving policy architecture that keeps revising and deepening the digital agenda. In contrast, many digital programmes in developing and transition economies remain one-off initiatives that lose momentum after the first phase of implementation. Kazakhstan's case is stronger precisely because its digital policy has been updated rather than abandoned[43].

The latest stage of this evolution is visible in the World Bank-supported Digital Acceleration for an Inclusive Economy (DARE) Project, approved in February 2024. The World Bank states that the project will provide high-quality and climate-resilient broadband access in underserved regions, with the goal of benefiting over one million people in rural areas. The same World Bank statement connects the project to Kazakhstan's 2025 National Development Plan and to the Digital Kazakhstan State Program, explicitly linking broadband expansion to innovation, digitalization, and a transition toward a more sustainable and competitive economic model. This is a very important point for your thesis: Kazakhstan's digital strategy has moved from building portals and governance systems toward reducing the digital divide and supporting broader structural competitiveness. That shift makes Kazakhstan especially useful as a benchmark for countries like Uzbekistan and Kyrgyzstan, where inclusion, access, and productivity remain central digital-development concerns[44].

What international indicators show

In the UN E-Government Development Index (EGDI) 2024, Kazakhstan ranked 24th with a score of 0.9009, while Uzbekistan ranked 63rd with 0.7999, and Kyrgyzstan ranked 78th with 0.7316. This means Kazakhstan belongs to the very high e-government group and is not just marginally ahead of the other two countries, but significantly more advanced in digital public-service delivery, telecommunication infrastructure, and human-capital readiness as reflected in the index methodology. Since public digitalization is one of the most visible outputs of state-led digital reform in Central Asia, the EGDI provides one of the strongest international justifications for choosing Kazakhstan as the benchmark country[45].

A second important benchmark is the Global Innovation Index (GII) 2024. In the WIPO rankings, Kazakhstan placed 78th, ahead of Uzbekistan at 83rd and Kyrgyzstan at 99th. The Kazakhstan country profile is especially useful because it shows not only the headline rank, but also the structure of Kazakhstan's innovation performance. WIPO identifies Kazakhstan's main strengths as government's online service (rank 8), utility models by origin per billion PPP GDP (rank 10), and e-participation [46](rank 15). The same profile also shows that Kazakhstan performs relatively well in ICTs overall and has a notable position in high-tech exports as a share of total trade (rank 36), even though weaknesses remain in software spending, scientific output, and R&D intensity. These details are important because they reveal the nature of Kazakhstan's benchmark status: the country is strongest where state digital capacity and administrative digitization matter most, while still lagging in deeper research and innovation inputs. That is exactly why it is a benchmark within Central Asia, not a fully advanced digital economy by global standards[47].

World Bank indicators reinforce this picture. On the World Bank's country data pages, internet use in Kazakhstan reached 93.39% of the population in 2024, compared with 90% in Uzbekistan and 92% in the Kyrgyz Republic. In terms of secure digital infrastructure, the contrast is even sharper: World Bank data show 6,914 secure internet servers per one million people in Kazakhstan in 2024, compared with 967 in Uzbekistan and 1,302 in the Kyrgyz Republic. These figures suggest that Kazakhstan's advantage is not only about having more internet users; it is also about having deeper digital infrastructure capable of supporting transactions, online services, and technology-intensive economic activity. In other words, Kazakhstan's lead is more visible in the quality and architecture of digitalization than in basic access alone[48].

Effect of digitalization on economic dynamics

The economic significance of digitalization in Kazakhstan becomes clearer when it is placed against the structure of the Kazakh economy. OECD notes that hydrocarbon and mineral resources remain the backbone of Kazakhstan's economy, with about 70% of total net FDI inflows going to the extraction sector between 2016 and 2020. The same OECD report states that the share of fossil fuels and the energy sector remained around 72% of goods exports since 2013, while the energy sector generated 17% of GDP in 2019. These data are crucial for understanding why digitalization matters so much in Kazakhstan. The country is not simply digitizing for convenience; it is trying to use digitalization as part of a longer-term diversification strategy that reduces excessive dependence on extractive growth. In this context, digitalization becomes a tool for changing the structure of the economy, not merely modernizing existing bureaucracy.

OECD explicitly supports this interpretation. It states that improving the framework conditions for private-sector digitalization can advance Kazakhstan's long-term diversification and growth agenda. The report also notes that during the first year of the COVID-19 pandemic, many Kazakh firms moved business operations online, which helped dampen the negative effect on the economy. This is an important transmission channel: digitalization increases resilience by allowing firms to continue transacting, communicating, and serving customers even under shock conditions. The broader World Bank framework in Digital Dividends explains that digital technologies can generate faster growth, more jobs, and better services, but only when they are supported by "analog complements" such as competition-friendly regulation, skills, and accountable institutions[49]. A second channel is public-sector efficiency. Kazakhstan's high EGDI position and WIPO strengths in online government service and e-participation indicate that digitalization has significantly improved the state's ability to deliver services, reduce administrative burden, and create more transparent interfaces between government, firms, and citizens. This point is also emphasized in the academic literature. Bokayev and co-authors argue that Kazakhstan's e-government trajectory should be judged not merely by the presence of online portals, but by how digital tools improve citizen interaction and service delivery[50]. Jussupova, Bokayev, and Zhussip similarly discuss digital government maturity as the stage at which technology begins to transform the logic of public administration rather than only digitize existing paperwork. In economic terms, such improvements reduce transaction costs, shorten administrative delays, and improve the business environment, all of which can raise productivity indirectly[51].

A third channel is innovation and technology adoption in firms and sectors. OECD's Central Asia review notes that Kazakhstan's digitalization agenda has included the use of 3D printing and the Industrial Internet of Things in industry. This matters because it connects digitalization to actual production processes, not just information access. Recent empirical studies on Central Asia also support the economic relevance of these channels. Shodiev, Turayey, and Shodiyev find a positive ICT-growth relationship in Central Asian countries. Kuldashaeva, Eshov, and Ismailova show that digital-economy variables, including ICT and e-commerce, are significant in explaining economic growth in the five Central Asian republics[52]. Yuldoshboy and co-authors conclude that ICT can be a sustained driver of economic growth in Central Asia, but only with complementary factors such as human capital and institutional strength[53]. Taken together, these studies do not prove that Kazakhstan's digitalization automatically causes every macroeconomic gain, but they do strongly support the argument that digitalization is a meaningful part of the region's growth model.

What lessons matter for Uzbekistan and Kyrgyzstan

The first lesson for Uzbekistan and Kyrgyzstan is that digitalization works best when it is integrated into a long-term economic-development strategy,

not treated as an isolated ICT programme. Kazakhstan's experience shows the value of continuity: a digital agenda beginning in 2013, a structured national programme in 2018, follow-up ecosystem tools such as Astana Hub and cybersecurity initiatives, and continued investment through DigitEL and the DARE project[54]. Uzbekistan is already moving in this direction. The World Bank's documentation on the Digital Uzbekistan 2030 strategy states that the Uzbek government plans to increase the size of the digital economy by at least 2.5 times, expand software outsourcing tenfold to USD 500 million, and bring digitalization of government workflow [55] to 90%. The lesson from Kazakhstan is that such ambitions need stable implementation architecture and not only headline targets.

The second lesson is that public digital services are necessary but not sufficient. Uzbekistan's OECD review stresses that the private sector has not yet fully harnessed the opportunities created by the government's digital strategy because firms still lack the institutional framework, awareness, and digital skills needed for effective uptake[56]. Kazakhstan's experience supports exactly this warning: even with strong e-government and better international rankings, private-sector digital adoption remains shallow in many segments. Kyrgyzstan's case makes this lesson even sharper. The World Bank's 2025 Country Economic Memorandum argues that weak productivity growth, human-capital skills gaps, and low-quality investment continue to hinder growth in the Kyrgyz Republic. For Kyrgyzstan, the Kazakh benchmark suggests that digital strategy must be tied to productivity, skills, and investment quality; otherwise digitalization risks remaining fragmented and low-impact.

The third lesson is that quality of infrastructure matters more than access alone. Kazakhstan's case shows that high internet-use rates do not automatically guarantee business digitalization if broadband quality, rural coverage, and security conditions are weak. This is particularly relevant for both Uzbekistan and Kyrgyzstan, where digital expansion is still uneven across territory and across firms. For Uzbekistan, the lesson is to move from expansion toward productive use. For Kyrgyzstan, the lesson is even more basic: digital transformation must be built on reliable connectivity, implementation capacity, and a supportive business environment. In both countries, Kazakhstan's experience suggests that rural connectivity, SME adoption, and digital skills deserve at least as much policy attention as e-government rankings.

The final lesson is methodological as well as policy-related: benchmarking should guide adaptation, not imitation. Kazakhstan's digital trajectory cannot simply be copied because Kazakhstan's scale, fiscal resources, administrative capacity, and international investment profile differ from those of Uzbekistan and especially Kyrgyzstan. But Kazakhstan does offer a useful regional demonstration of what matters most: policy continuity, integration of digitalization with diversification goals, attention to digital government and business ecosystems simultaneously, and the use of international indicators to monitor progress. For that reason, Kazakhstan remains the most appropriate benchmark in this thesis, while Uzbekistan and Kyrgyzstan remain indispensable comparative cases that reveal where digital transformation still needs deeper institutional and productive foundations[57].

2.2. Comparative benchmarking of digital transformation and high-technology adoption in Kazakhstan, Uzbekistan and Kyrgyzstan: analytical framework, country selection and cross-country differences

Digitalization as a driver of economic growth

Digitalization has become one of the most important structural determinants of modern economic growth, because it affects not only access to information and communication technologies, but also innovation capability, firm productivity, public administration efficiency, and the diffusion of advanced technologies across sectors. For this reason, the present section does not reduce digital transformation to a single ICT variable. Instead, it examines digitalization as a multidimensional process involving digital infrastructure, digital public services, business digital adoption, innovation capacity, and selected indicators of high-technology activity. Such an approach is consistent with the OECD measurement framework, which stresses that digital transformation must be assessed through indicators ranging from education and innovation to trade, economic outcomes, and public policy performance[58]. From a methodological perspective, this approach is especially important for Central Asia, where digital transformation is uneven and where the economic effects of digitalization depend not only on internet access, but also on state capacity, quality of institutions, business readiness, and the ability to translate digital access into productive and innovative activity.

The UN ESCAP background paper on North and Central Asia emphasizes that digital transformation in the subregion cannot be understood only through connectivity indicators, because sectoral applications in e-commerce, finance, agriculture, supply chains, and logistics are equally important for sustainable development[59]. In the same way, OECD work on Central Asia argues that digital transformation should be studied both through infrastructure and through the extent to which firms, especially SMEs, actually adopt digital tools and integrate them into business activity. International academic literature also supports a broader interpretation of the digitalization-growth nexus. In the European Union, Magoutas, Chaideftou, Skandali, and Chountalas show that ICT development is positively associated with GDP performance and that the contribution of digital transformation is strengthened when it is combined with e-government progress and human-capital enhancement[60]. In another recent comparative study, Correa and Esquivias conclude that digitalization, education, and institutional quality positively influence economic growth in both Sub-Saharan Africa and the Middle East, while the strength of the effect depends on differences in institutional development and learning capacity. These studies are important for the present thesis because they demonstrate that the economic impact of digitalization is rarely direct or automatic; rather, it is mediated by complementary factors such as education, governance, innovation, and policy design[61].

A growing number of studies have already addressed this issue in the Central Asian context, although the literature remains relatively limited compared with Europe or East Asia. Shodiev, Turayev, and Shodiyev examined the ICT-growth nexus for Central Asian countries over 2000-2018 and found that ICT variables such as fixed broadband subscriptions and internet use positively affected GDP per capita[62]. Kuldasheva, Eshov, and Ismailova studied five Central Asian republics over 2000-2022 and concluded that e-commerce and ICT had a significant positive relationship with economic growth in the region[63]. More recently, Yuldoshboy and co-authors investigated whether ICT functions as a sustained driver of growth in four Central Asian economies over 2000-2022 within an extended endogenous growth framework, while Kurmanov and colleagues reported a significant positive impact of ICT on economic growth, especially in Kazakhstan and Uzbekistan[64]. Together, these studies confirm that the digitalization-growth relationship in Central Asia is already an established research field, but one that still requires more focused comparative analysis of selected countries and more precise identification of benchmark cases. Against this background, the choice of Kazakhstan, Uzbekistan, and Kyrgyzstan for comparative benchmarking is methodologically justified. First, all three countries belong to the same subregion and share several structural similarities: they are post-Soviet economies, landlocked states, and transition systems in which the state continues to play a major role in economic modernization and public-sector reform. Second, all three governments have adopted formal digital transformation strategies, which makes them comparable at the policy level. Third, despite these similarities, the three countries differ significantly in digital maturity, innovation ranking, implementation capacity, and business adoption of digital tools. This combination of common regional characteristics and unequal digital outcomes makes them especially suitable for comparative study. In other words, they are similar enough to be compared within one regional framework, but different enough to reveal meaningful contrasts in the relationship between digitalization, high-technology adoption, and economic growth.

Within this group, Kazakhstan can be treated as the benchmark economy. OECD notes that Kazakhstan has been developing an ambitious digitalization agenda since 2013, while its dedicated Digital Kazakhstan programme operated as a five-year plan from 2018 to 2022[65]. According to OECD's regional review, Kazakhstan's strategy covered broadband expansion, high-speed access, workforce upskilling, one-stop business services,

cybersecurity, SME-targeted support, digital finance, digital payments, and e-commerce, giving it the broadest and most institutionally integrated digital policy package among the three selected countries. The same OECD source stresses that the programme encouraged the use of technologies such as 3D printing and the Industrial Internet of Things (IoT) in the extractive and manufacturing sectors, which is especially relevant for the "high technologies" component of the present thesis. This means that Kazakhstan is not merely a country with better internet access; it is a case where digital transformation has been linked more clearly to industrial modernization and public-sector digitization. Kyrgyzstan is an equally important comparative case, but for different reasons. OECD's regional review shows that the Digital Kyrgyzstan 2019-2023 programme aimed to create the basic foundations of digital transformation through regional backbone networks, fibre-optic expansion, and digital literacy development. The same source notes that the government launched the Tunduk unified identification and authentication system and accelerated online public services during the COVID-19 period[66].

International rankings confirm this more constrained position. In the Global Innovation Index 2024, Kyrgyzstan ranked 99th, which places it well behind the two other countries selected for this thesis. World Bank data show that internet use still reached approximately 92% in 2024, while mobile cellular subscriptions were about 109 per 100 people in 2023. However, fixed broadband subscriptions in the Kyrgyz Republic were only about 6.4 per 100 people in 2024, substantially below Uzbekistan's level and also below Kazakhstan's low-teen range reported in the World Bank DataBank. This mixed picture indicates that Kyrgyzstan has achieved significant basic digital access, but has not yet converted that access into a comparable level of institutional digital maturity, innovation depth, or technology-intensive economic performance[67]. Uzbekistan, by contrast, represents a fast reformer rather than an already consolidated benchmark. OECD describes Uzbekistan as having embarked on an ambitious reform agenda after 2016 and notes that in 2020 the country adopted the Digital Uzbekistan 2030 Strategy, aimed at integrating digital technologies into public services, infrastructure, the economy, and education. A particularly important result of the comparative assessment is that public-sector digitalization is generally stronger than firm-level digitalization across Central Asia. OECD's regional note points out that, based on World Bank Enterprise Survey rounds between 2017 and 2021, the percentage of Central Asian firms having their own website was below peer-group averages. The gap between upper-middle-income comparators and Kazakh firms was 14% points, while the gap between lower-middle-income comparators and Uzbek firms was 19.2% points. OECD also notes that Kazakhstan's internal gap between large firms and small firms in website use was 40.8% points, which suggests that even in the regional leader, digitalization remains uneven across the business sector. This observation is highly relevant for the present thesis, because it implies that growth effects from digitalization in Central Asia may depend not only on internet access or e-government development, but also on whether digital tools diffuse beyond the public sphere into firms, SMEs, manufacturing, trade, and service activities[68].

Therefore, the main comparative conclusion of this section is not simply that Kazakhstan is "more digitalized" than Uzbekistan and Kyrgyzstan. The more precise conclusion is that the three countries represent different stages and qualities of digital transformation. Kazakhstan combines stronger digital government, broader strategic coverage, higher innovation standing, stronger secure-server infrastructure, and a higher share of high-technology exports, which justifies its role as a benchmark case. In sum, the comparative benchmarking of Kazakhstan, Uzbekistan, and Kyrgyzstan confirms that Kazakhstan is the most appropriate reference case for this thesis. At the same time, the comparison shows that benchmarking should not be interpreted as imitation in a mechanical sense. Kazakhstan's experience is valuable because it demonstrates how a broader and more institutionally coherent digital strategy can be associated with better outcomes in e-government, innovation, and selected technology indicators. Uzbekistan and Kyrgyzstan remain indispensable to the analysis because they reveal different reform trajectories, different implementation capacities, and different distances from the benchmark. This comparative logic creates the foundation for the next parts of the chapter: first, a more focused discussion of Kazakhstan's benchmark role, and then the econometric examination of how digitalization and high-technology indicators affect economic growth in the three selected countries.

2.3. Econometric analysis of the effect of digitalization on economic development of Kazakhstan, Uzbekistan, and Kyrgyzstan (2004-2023 years)

Objectives and tasks of the study

The main purpose of this econometric analysis is to empirically assess the relationship between the impact of digitalization on economic growth in Kazakhstan, Uzbekistan and Kyrgyzstan between 2004 and 2023. In this work, GDP growth is used as the dependent variable, while internet use, mobile subscriptions, broadband development are included as explanatory factors. The analysis further aims to compare pooled OLS, fixed effects, and random effects estimations and to identify the most appropriate model through the Hausman test. The study primarily focuses on understanding how it-related variables affect per capita GDP, which serves as a variable for economic development.

Data and variables

The econometric analysis is based on an annual panel dataset for Kazakhstan, Uzbekistan, and Kyrgyzstan covering the period 2004-2023. The dataset was compiled primarily from the World Bank's World Development Indicators (WDI), while the digitalization indicators used in WDI are largely based on the International Telecommunication Union (ITU) database.

- Dependent variable:
- 7. gdpcap: GDP growth (annual %)
- Independent variable:
- 8. inuser: individuals using the internet (% of population)
 - 9. broad: fixed broadband subscriptions (per 100 people)
 - 10. mobsub: mobile cellular subscriptions (per 100 people)
 - 11. trade: trade (%of GDP)
 - 12. inflat: inflation, GDP deflator (annual %)

These indicators were selected based on their theoretical relevance and availability of information and normalized to ensure compliance if necessary.

Figure 2.3.1. Types of variables

Source: Prepared by the author

The data set shown is a panel data frame containing 60 observations over 20 years (2004-2023) across 3 countries. Each row represents a country-Year combination, and the variables include key economic and numerical indicators.

The main variable is GDP per capita, which indicates the average per capita income. Other variables include:

- 13. gdpcap: GDP growth (annual %)
- 14. inuser: individuals using the internet (% of population)
- 15. broad: fixed broadband subscriptions (per 100 people)
- 16. mobsub: mobile cellular subscriptions (per 100 people)
- 17. trade: trade (%of GDP)
- 18. inflat: inflation, GDP deflator (annual %)

Figure 2.3.2. Information about variables

Source: Prepared by the author

This summary in Figure 2.3.2 provides a descriptive view of the cleaned panel data set. The data includes observations from 3 countries, each presented over a 20-year period from 2004 to 2023. The average GDP growth rate across all three countries is approximately 3.6%, with values ranging from -9% (Kyrgyzstan, 2020) to +9.8% (Kazakhstan, 2006). The distribution shows moderate variability, reflecting differences in economic cycles and external shocks such as the 2008-2009 global financial crisis and the 2020 COVID-19 pandemic. Internet penetration grew substantially across the panel, with an average of around 46% and a wide range (2.6%-92.9%), reflecting the rapid digitalisation of these economies over 2004-2023. Average mobile subscriptions per 100 people stand at approximately 91, with Kazakhstan reaching the highest saturation levels. Average fixed broadband subscriptions are around 7.2 per 100 people, indicating that fixed broadband infrastructure remains underdeveloped compared to mobile connectivity. Kyrgyzstan shows significantly higher trade-to-GDP ratios (up to 146%), while Uzbekistan historically maintained lower trade openness. Inflation exhibits the largest spread (1.8%-48.6%), with Uzbekistan recording notably high inflation during the early years of the panel.

Figure 2.3.3. Boxplot analysis by country

Source: Prepared by the author

Figure 2.3.3 demonstrates the distribution, median values, dispersion, and possible outliers of the main variables across the Kazakhstan, Kyrgyzstan, Uzbekistan . Kyrgyzstan exhibits the widest spread and most outliers, indicating higher economic volatility. Kazakhstan shows more stable growth at the upper end. Uzbekistan maintains moderate but consistent growth. Kazakhstan leads with the highest and most consistent internet penetration, followed by Uzbekistan. Kyrgyzstan shows lower and more dispersed values. All three countries show high mobile penetration, with Kazakhstan being the highest. Outliers suggest supersaturation (values above 100) typical of multiple-SIM users. All three countries have low fixed broadband values, but Kazakhstan is again the leader. Uzbekistan has shown recent growth. Kyrgyzstan's trade-to-GDP ratio is substantially higher and more volatile - reflecting its reliance on re-export trade and smaller domestic market. Uzbekistan shows significantly higher and more variable inflation, particularly in the early years. Kazakhstan and Kyrgyzstan are more moderate.

Figure 2.3.4. Boxplot analysis by year

Source: Prepared by the author

Figure 2.3.4 shows boxplot analysis by year. GDP growth shows clear negative shocks in 2009 (global financial crisis) and 2020 (COVID-19 pandemic), visible as drops across all countries. "Internet users" xhibits a steady upward trend every year, confirming rapid digitalisation across the region from 2004 to 2023. "Mobile subscriptions" grew rapidly through 2011-2012 before stabilising, as market saturation was reached. "Fixed broadband" shows slow but consistent growth, especially accelerating post 2015. "Trade" shows a declining trend post-2010, particularly for Kyrgyzstan, and disruption in 2020. "Inflation" was notably high and variable in the early years (2004-2008), stabilised mid-period, and spiked again in 2022 due to global commodity shocks.

Figure 2.3.5. GDP growth

Source: Prepared by the author

All three economies experienced negative growth in 2020 due to the COVID-19 pandemic. Kazakhstan was most severely affected (-3.9%). Kyrgyzstan showed the most volatility across the period, with a deep contraction in 2020 (-9%). Uzbekistan maintained more resilient growth throughout.

Figure 2.3.6. Internet users (% of population)

Source: Prepared by the author

All three countries show a clear upward trend. Kazakhstan was consistently ahead. By 2023, all three exceed 88% penetration, indicating near-universal access.

Figure 2.3.7. Mobile subscriptions (Per 100 people)

Source: Prepared by the author

Figure 2.3.7 shows a clear long-term expansion of mobile subscriptions in Kazakhstan, Kyrgyzstan, and Uzbekistan over the study period. Kazakhstan demonstrates the highest level throughout most of the period, with especially rapid growth until the early 2010s, after which the indicator stabilizes and slightly declines. Kyrgyzstan also records strong growth, reaching a peak in the late 2010s before moderating afterward. Uzbekistan starts from the lowest base, but shows a steady upward trend and catches up significantly after 2018-2019. Overall, the graph indicates that mobile communication infrastructure expanded substantially in all three countries, although the pace and timing of growth differed across them.

Figure 2.3.8. Inflation rate

Source: Prepared by the author

Figure 2.3.9. Fixed broadband subscriptions

Source: Prepared by the author

Figure 2.3.9 shows that fixed broadband subscriptions increased in all three countries, but with different trajectories. Kazakhstan was the early leader, demonstrating steady and relatively stable growth from the late 2000s onward. Kyrgyzstan showed the weakest performance, with only gradual improvement and the lowest subscription levels throughout the period. Uzbekistan began from a very low base, but after the mid-2010s it experienced the fastest growth and eventually exceeded the other two countries. This indicates that broadband infrastructure developed earliest in Kazakhstan, most slowly in Kyrgyzstan, and most dynamically in Uzbekistan in the later years.

Figure 2.3.10. Trade (% of GDP)

Source: Prepared by the author

Figure 2.3.10 illustrates that Kyrgyzstan had the highest trade-to-GDP ratio during most of the study period, although this indicator was highly volatile. Kazakhstan and Uzbekistan both started from low levels of trade openness, but their ratios increased significantly over time. Kazakhstan experienced a sharper early rise and later stabilization, while Uzbekistan showed a steadier and more gradual increase. In general, the figure reflects stronger and more

sustained trade integration in all three countries, with Kyrgyzstan remaining the most externally open economy relative to GDP.

Scatter plot analysis of the relationship between GDP growth and explanatory variables

Figure 2.3.11. Internet users vs. GDP growth

Source: Prepared by the author

Figure 2.3.11 shows a weak negative relationship between internet use and GDP growth. Although the fitted line slopes downward, the observations are widely scattered, suggesting that the bivariate relationship is not strong. This means that internet use alone cannot explain changes in GDP growth and that further econometric analysis is necessary.

Figure 2.3.12. Fixed broadband subscription vs. GDP growth

Source: Prepared by the author

Figure 2.3.12 suggests a weak negative relationship between fixed broadband subscriptions and GDP growth. Although the fitted trend line slopes downward, the broad dispersion of observations shows that the bivariate relationship is not strong. Therefore, the graph should be treated only as a preliminary visual assessment, while the final conclusion should be based on panel regression results.

Figure 2.3.13. Correlation matrix

Source: Prepared by the author

Figure 2.3.13 reveals several important relationships:

- GDP Growth and Internet Users: weak negative correlation (≈ -0.21), which may reflect the time-lag structure between digitalisation and growth effects, or the influence of other confounders.
- GDP Growth and Inflation: moderate negative correlation (≈ -0.28), consistent with economic theory that high inflation is detrimental to growth.
- GDP Growth and Trade (% of GDP): negative correlation (≈ -0.30), largely driven by Kyrgyzstan's high trade ratio co-occurring with volatile and sometimes negative growth.
- Among predictors: Internet Users, Mobile Subscriptions, and Fixed Broadband are highly intercorrelated (0.90-0.96), indicating multicollinearity. Including all three simultaneously in a regression may distort individual coefficient estimates.

Figure 2.3.14. Results of the pooled OLS regression model

Source: Prepared by the author

The pooled OLS estimation examines the effect of digitalization indicators and selected macroeconomic control variables on GDP growth in Kazakhstan, Uzbekistan, and Kyrgyzstan over the period 2004-2023. The model is based on a balanced panel with 3 countries, 20 years, and 60 observations.

The model was estimated based on the following equation:

$$\text{GDP_Growth} = 3.92 - 0.02 \cdot \text{Internet_Users} - 0.01 \cdot \text{Mobile_Subscriptions} + 0.06 \cdot \text{Fixed_Broadband} - 0.01 \cdot \text{Trade_GDP} + 0.14 \cdot \text{Inflation}$$

Main results:

The regression results show that inflation is the only explanatory variable that is statistically significant in the pooled model. Its coefficient is 0.1427, with a p-value of 0.0081, which means that a one-unit increase in inflation is associated with an increase of about 0.143 percentage points in GDP growth, holding other factors constant. This positive relationship may reflect the specific macroeconomic conditions of the selected countries during the study period, where moderate inflation may have accompanied periods of higher economic activity. However, the explanatory power of the model is moderate: the R-squared is 0.3142, which means that about 31.4% of the variation in GDP growth is explained by the included regressors. The adjusted R-squared of 0.2507 suggests that the explanatory strength decreases somewhat after accounting for the number of variables included in the model.

Figure 2.3.15. Fixed effects model

Source: Prepared by the author

This output presents a model of fixed effects (within) that governs unobserved country-specific factors by allowing each country to have its own barrier. This approach helps determine the actual impact of explanatory variables on GDP per capita over time within each country. The estimated R-squared is 0.2607 and the adjusted R-squared is 0.1612, which means that after accounting for unobserved country-specific effects, approximately 26% of the variation in GDP growth is explained by the included explanatory variables. Although this explanatory power is lower than in the pooled OLS model, such a result is expected because the Fixed Effects specification applies a stricter estimation approach and concentrates only on within-country variation over time. The overall model remains statistically significant, as indicated by the F-statistic of 3.6669 with a p-value of 0.00645, which confirms that the regressors are jointly relevant in explaining GDP growth.

Among the explanatory variables, inflation is the only factor that remains statistically significant in the Fixed Effects model. Its coefficient is 0.1377 with a p-value of 0.0163, indicating that a one-unit increase in inflation is associated with an increase of about 0.138 percentage points in GDP growth, holding other variables constant. This suggests that, within the selected countries, periods of higher inflation were accompanied by higher recorded GDP growth, possibly reflecting the broader macroeconomic dynamics of developing and transition economies during the study period. The coefficients of the main digitalization variables are not statistically significant. Internet users has a negative coefficient ($\beta = -0.0276$, $p = 0.3322$), which suggests a negative relationship with GDP growth, but the effect is not statistically reliable. Mobile subscriptions also has a negative and insignificant coefficient ($\beta = -0.0124$, $p = 0.3868$). Fixed broadband shows a very small negative coefficient ($\beta = -0.0131$, $p = 0.9243$), which is far from statistical significance. Similarly, trade openness has a positive coefficient ($\beta = 0.0119$, $p = 0.6085$), but its effect is also statistically insignificant. These results indicate that, once country-specific effects are taken into account, the selected digitalization indicators and trade openness do not show an independent statistically significant impact on GDP growth in this specification.

Figure 2.3.16. Random effects model

Source: Prepared by the author

The Random Effects model evaluates the relationship between digitalization indicators, selected macroeconomic control variables, and GDP growth in Kazakhstan, Uzbekistan, and Kyrgyzstan over the period 2004-2023. The estimation is based on a balanced panel consisting of 3 countries, 20 years, and 60 observations. Unlike the Fixed Effects model, the Random Effects specification assumes that country-specific effects are random and uncorrelated with the explanatory variables, which makes it possible to use both within-country and between-country variation in the estimation.

The model has an R-squared of 0.2795 and an adjusted R-squared of 0.2128, which means that approximately 28% of the variation in GDP growth is explained by the included regressors. Although the explanatory power is moderate, the overall model is statistically significant. This is confirmed by the Chi-square statistic of 20.9461 with a p-value of 0.000829, which indicates that the explanatory variables are jointly relevant in explaining GDP growth.

Important result:

Among all explanatory variables, inflation is the only variable that is statistically significant in the Random Effects model. Its coefficient is 0.1444 with a p-value of 0.0061, which implies that a one-unit increase in inflation is associated with an increase of about 0.144 percentage points in GDP growth, holding other variables constant. This result is consistent with the pooled OLS and Fixed Effects models, where inflation also appeared as the only statistically significant determinant. The positive coefficient suggests that, in the selected countries, periods of higher inflation were associated with periods of stronger economic growth, possibly reflecting the macroeconomic conditions of developing economies during the study period.

The remaining explanatory variables do not show statistical significance. Internet users has a negative coefficient ($\beta = -0.0296$, $p = 0.2868$), indicating a weak negative association with GDP growth, but the effect is not statistically reliable. Mobile subscriptions also has a negative and insignificant coefficient ($\beta = -0.0100$, $p = 0.4571$). Fixed broadband has a positive coefficient ($\beta = 0.0506$, $p = 0.6365$), suggesting a possible positive contribution to growth, but the relationship is not statistically significant. Trade openness likewise has a negative and insignificant coefficient ($\beta = -0.0039$, $p = 0.7661$), meaning that no clear independent effect on GDP growth can be identified in this specification.

Figure 2.3.17. F-test

Source: Prepared by the author

The F-test for individual effects was conducted to determine whether the Fixed Effects model is more appropriate than the pooled OLS specification.

Following results:

$F = 0.53254$

Degrees of freedom: $df_1 = 2$, $df_2 = 52$

p-value = 0.5903

The test produced an F-statistic of 0.5325 with a p-value of 0.5903. Since the p-value is greater than 0.05, the null hypothesis cannot be rejected. This indicates that individual country-specific effects are not statistically significant in the model. Therefore, the Fixed Effects specification does not provide a significant improvement over the pooled OLS model.

Figure 2.3.18. Breusch-Pagan test

Source: Prepared by the author

The Breusch-Pagan Lagrange Multiplier test was applied to determine whether the Random Effects model is more appropriate than the pooled OLS specification.

Following results:

Chi-square = 1.2179

Degree of freedom = 1

p-value = 0.2698

The test produced a chi-square statistic of 1.2179 with a p-value of 0.2698. Since the p-value exceeds the 5% significance level, the null hypothesis cannot be rejected. This indicates that random country-specific effects are not statistically significant. Therefore, the Random Effects model does not provide a significant improvement over the pooled OLS model.

Figure 2.3.19. Hausman test

Source: Prepared by the author

The Hausman test is used to compare the Fixed Effects and Random Effects models and to determine which of the two is more appropriate for the panel dataset.

The result shows the following:

1. Chi-square = 0.77088

2. Degrees of freedom = 5

3. p-value = 0.9789

Since the p-value is much greater than 0.05, the null hypothesis cannot be rejected. This means that there is no systematic difference between the Fixed Effects and Random Effects estimators, and therefore the Random Effects model can be considered consistent and efficient in comparison with the Fixed Effects model.

However, in the case of this study, the Hausman test should not be interpreted alone. The earlier F-test showed that the Fixed Effects model is not significantly better than pooled OLS, and the Breusch-Pagan LM test also showed that the Random Effects model is not significantly better than pooled OLS. Therefore, although the Hausman test formally favors the Random Effects model over the Fixed Effects model, the overall diagnostic results indicate that pooled OLS remains the most appropriate specification for this dataset.

During my analysis, I tried to understand how digitalization and selected macroeconomic factors are related to GDP growth in Kazakhstan, Uzbekistan, and Kyrgyzstan. After estimating the pooled OLS, Fixed Effects, and Random Effects models, I found that the results were quite similar across all specifications. The most important finding was that inflation was the only variable that remained statistically significant. In the Random Effects model, for example, its coefficient was about 0.144, with a p-value of 0.006, which shows a positive and significant relationship with GDP growth.

I also expected the main digitalization indicators - internet users, mobile subscriptions, and fixed broadband - to have a clearer effect on economic growth. However, in my model, these variables were not statistically significant. This does not necessarily mean that digitalization has no economic importance. Rather, it suggests that its effect may be more indirect and may depend on other supporting conditions such as institutions, investment, and human capital.

In summary, the Hausman test suggests that the Random Effects model is preferable to the Fixed Effects model, but when all diagnostic tests are considered together, the pooled OLS model is the most suitable final model for this research.

Conclusion for the 2.3 part

In conclusion, the econometric analysis examined the relationship between digitalization indicators and GDP growth in Kazakhstan, Uzbekistan, and

Kyrgyzstan over the period 2004-2023 using panel data methods. The pooled OLS, Fixed Effects, and Random Effects models produced broadly similar results, with inflation emerging as the only statistically significant variable across the estimated specifications. By contrast, the digitalization indicators - internet users, mobile subscriptions, and fixed broadband subscriptions - did not show statistically significant direct effects on GDP growth in the estimated models. The model selection tests further indicated that neither fixed nor random country-specific effects were sufficiently strong, and therefore the pooled OLS model was identified as the most appropriate final specification for this dataset.

Overall, the results suggest that in the selected Central Asian countries, digitalization may not yet exert a strong short-run measurable effect on economic growth in direct linear form, while macroeconomic conditions appear to play a more immediate role. At the same time, the descriptive and comparative analysis indicates that digitalization remains an important long-term structural factor for economic modernization and future development.

CHAPTER III: ADAPTATION OF INTERNATIONAL EXPERIENCE AND STRATEGIC SUGGESTIONS FOR UZBEKISTAN

3.1 Present situation of digitalization and economic evolution in Uzbekistan

Among the three Central Asian economies examined in this thesis, Uzbekistan occupies a strategically distinctive position. It is neither the established benchmark that Kazakhstan represents, nor the structurally constrained laggard that Kyrgyzstan illustrates. Instead, Uzbekistan is best understood as a fast reformer whose digital trajectory has accelerated sharply since 2016-2017, but whose reform process remains uneven, institutionally incomplete, and exposed to structural bottlenecks that could prevent digitalization from generating its expected economic dividends. Understanding where Uzbekistan stands today - its strategic strengths, its progress on key indicators, and its remaining weaknesses - is essential before any meaningful policy recommendation can be formulated.

In macroeconomic terms, Uzbekistan has demonstrated remarkable resilience and dynamism over the study period. GDP growth averaged approximately 3.6% across the three-country panel for 2004-2023, and Uzbekistan consistently recorded stronger performance than Kyrgyzstan, particularly in years of external shocks. In the first quarter of 2024, Uzbekistan's economy grew by 6.2% year-on-year, reflecting the continued impact of structural reforms launched since 2017 and sustained government investment in infrastructure and human capital.[69] With a GDP of approximately USD 114.97 billion and a population of over 37 million in 2024, Uzbekistan is the most populous economy in Central Asia, giving it both a large domestic market and a substantial human capital base that, if effectively harnessed through digital skills development, could become a significant growth driver. The strategic architecture of Uzbekistan's digital transformation rests on several interconnected policy pillars. The foundational document is the Digital Uzbekistan 2030 Strategy, formally adopted in 2020, which aims to integrate digital technologies across public services, infrastructure, the broader economy, and education. The strategy set ambitious headline targets, including increasing the size of the digital economy by at least 2.5 times, expanding software outsourcing tenfold to USD 500 million, bringing the digitalization of government workflow to 90%, and placing Uzbekistan among the top 30 countries globally in the UN E-Government Development Index by 2030.

This digital strategy does not exist in isolation. It is embedded within the broader Uzbekistan-2030 National Development Strategy, approved in September 2023, which sets 100 measurable national goals across five key areas - sustainable economic growth, the rule of law, environmental protection, a just state, and a peaceful society - and which explicitly ties human capital, science spending, and innovation performance to the country's aspiration of reaching upper-middle-income status by the end of the decade.[70] In addition, the government adopted in October 2024 a dedicated Artificial Intelligence Development Strategy until 2030, which allocated USD 50 million for AI infrastructure, set a target of creating a national AI model, and committed to training one million specialists in AI-related fields.[71] The adoption of this AI-specific strategy, complementing the broader digital agenda, indicates that Uzbekistan's policy framework has moved beyond first-generation e-government ambitions toward a more sophisticated orientation focused on deep digital transformation, high technologies, and knowledge-intensive growth.

In terms of digital infrastructure, Uzbekistan has achieved substantial progress, although notable quality gaps persist. Internet penetration reached approximately 90% of the population by 2024, bringing the country to near-parity with Kazakhstan (93.39%) and Kyrgyzstan (92%) in terms of basic access.[72] Mobile broadband coverage, according to GSMA data for 2025, extended to 99% of all settlements in the country, while 5G services were deployed in all regional centres by 2024.[73] These are significant headline achievements. However, the quality and depth of connectivity remain more limited. As of 2024, the World Bank reported only 967 secure internet servers per one million people in Uzbekistan, compared with 6,914 in Kazakhstan - a gap of more than sevenfold that reveals the structural shallowness of Uzbekistan's digital infrastructure relative to its benchmark neighbour.

Position of Uzbekistan

Uzbekistan's most internationally visible digital achievement has been in the area of e-government. In the UN E-Government Development Index (EGDI) 2024, the country ranked 63rd with a score of 0.7999, rising from 69th place in 2022 and exceeding the intermediate target of 0.75 EGDI set for 2025 within the Digital Uzbekistan 2030 Strategy ahead of schedule.[74] These gains reflected genuine progress in digital public-service delivery: the my.gov.uz portal now offers over 560 e-services, and the government's 2023-2024 Digital Government and Digital Region Programs outlined 300 priority digitalization projects, covering public administration digitization, real-economy sector transformation, and regional digital programmes.[75] The World Bank's GovTech Maturity Index additionally ranked Uzbekistan 43rd out of 198 countries in 2022, placing it in the highest maturity group (Group A), and rated the country among the world's top four performers on digital skills and innovation in public services.[76] Uzbekistan's IT sector and innovation ecosystem have also expanded rapidly, though from a modest base. ICT exports surged from a negligible USD 600,000 in 2017 to USD 140 million in 2022, and then to USD 619.74 million in 2024 - a 40% increase in a single year - reflecting the growing international competitiveness of the country's technology service sector.[77] The share of the ICT sector in GDP grew from approximately 2% in 2018 to approximately 5% by 2023,[78] with some estimates placing the digital sector's contribution even higher when broader digital economy activities are included.

Picture 3.1.1. ICT: Service export of Uzbekistan (2013-2024)

Source: HYPERLINK "<https://www.ceicdata.com/en/uzbekistan/telecommunication/uz-ict-service-exports-bop>"

<https://www.ceicdata.com/en/uzbekistan/telecommunication/uz-ict-service-exports-bop>

The institutional centrepiece of this innovation expansion is IT Park Uzbekistan, established in 2019 under the Ministry of Digital Technologies, which by 2025 supported over 2,800 companies - including more than 730 foreign firms - and offered zero-tax treatment for resident technology companies alongside preferential taxation for their employees.[79] In innovation rankings, Uzbekistan placed 83rd in the WIPO Global Innovation Index (GII) 2024, ahead of Kyrgyzstan (99th) but behind Kazakhstan (78th), and for the third consecutive year maintained its status as a GII innovation overperformer relative to its income level.[80] Despite these gains, Uzbekistan's digital transformation faces three interrelated structural challenges that define its strategic problem and distinguish it from Kazakhstan's more consolidated digital trajectory. The first and most consequential challenge is the disconnect between digital access and digital use, particularly at the firm level. OECD's 2023 review of digital skills for private-sector competitiveness in Uzbekistan concluded explicitly that, while the country had made significant progress in internet access, quality, and affordability, "digital uptake among firms

remains low" and that "the lack of digital skills is a limiting factor affecting the digital transformation of firms."

OECD's analysis identifies three root causes for low firm-level digital adoption: gaps in the institutional framework to support SME digitalization, a lack of awareness among firms regarding the benefits of digital transformation, and limited digital skills among both managers and employees. These are not peripheral problems. SMEs account for the majority of employment in Uzbekistan and a significant share of GDP, yet their capacity to benefit from the government's digital infrastructure investment remains severely constrained by the absence of coordinated, sector-neutral, and SME-friendly support mechanisms. The existing digital tool provision is scattered across multiple agencies, making it difficult for small firms to identify and access the services available to them.

The second structural challenge concerns investment and regulatory environment for digital sectors. OECD's 2025 investment roadmap for Uzbekistan notes that Uzbekistan's data localisation requirements - which mandate that **personal data of citizens must be stored within the country** - while generating some short-term data-centre investment, impose higher operational costs on firms, deter market entry by foreign cloud providers, and reduce access by domestic firms to global digital platforms, thereby limiting innovation spillovers.[81] The same OECD report observes that, unlike many OECD and non-OECD economies that experienced enhanced digital foreign direct investment following the COVID-19 pandemic, Uzbekistan witnessed a diverging trend, with digital FDI flows not keeping pace with the global surge.

The third structural challenge is regional and demographic unevenness. Urban-rural and inter-regional disparities in digital access, skills, and economic opportunity remain significant in Uzbekistan. Approximately half of the population of over 37 million lives in rural areas, where infrastructure quality is lower, digital literacy is weaker, and the distance from digital service hubs is greater. Agriculture still accounts for approximately 18% of GDP, and a large proportion of the labour force is engaged in activities with limited digital intensity. The risk, therefore, is that the benefits of digital transformation will concentrate in Tashkent and other urban centres, while the rural majority and micro-enterprises in traditional sectors remain largely outside the digital economy.

For Uzbekistan, the implication is precise: the country has invested heavily in building digital access infrastructure and e-government capacity, but has not yet generated the complementary conditions - digital skills at scale, firm-level adoption, innovation-enabling regulation, and deep broadband quality - needed for digitalization to produce statistically detectable effects on aggregate growth. This gap between infrastructure presence and productive digital use is the central strategic problem that Chapter III recommendations must address.

3.2. Significance of international experience and strategic advices and their expected economic outcomes to Uzbekistan

The comparative analysis developed in Chapter II established that Kazakhstan's digital transformation advantage over Uzbekistan rests not simply on higher internet penetration - which is now broadly similar across the three countries - but on three more substantive dimensions: the depth and quality of digital infrastructure, the sustained institutional architecture of digital policy over more than a decade, and the closer linkage between digitalization and broader economic diversification goals. Each of these dimensions offers a specific lesson for Uzbekistan, provided that the lesson is adapted to Uzbekistan's own structural conditions rather than mechanically copied from a country with significantly greater fiscal and administrative resources.

What is transferable from Kazakhstan's experience, and what is not, is therefore the analytical starting point for this section. Kazakhstan's institutional continuity - a digital agenda beginning in 2013, a five-year Digital Kazakhstan programme from 2018, the launch of Astana Hub, and DigitEL in 2022 - demonstrates that the primary prerequisite for digital transformation to produce economic results is policy consistency rather than policy ambition.[82] Uzbekistan has matched Kazakhstan's ambition level but not yet its implementation consistency. What is less directly transferable is Kazakhstan's approach to heavy state-led industrial digitalization in the extractive sector, which relied on a resource-revenue base and investment profile that Uzbekistan does not possess. Uzbekistan's comparative advantage lies in its large young population, lower labour costs, and the potential for ICT services exports and SME-led digital productivity gains - a different economic logic that requires a different digital strategy emphasis.

OECD's assessment of digital skills for private-sector competitiveness in Uzbekistan identifies the institutional gap clearly: there is no specialized agency responsible for providing structured digital literacy tools and adoption support to SMEs, existing incentives are concentrated in IT Park and IT- and BPO-sector priorities rather than being economy-wide, and regulatory barriers around personal data protection discourage even digitally capable firms from investing in digital transformation. In response, Uzbekistan should establish a dedicated one-stop-shop for SME digital transformation - analogous to the SME digital support frameworks that OECD countries have developed - that consolidates financial incentives, training programmes, certification schemes, and advisory services under a single institutional roof accessible to firms across all sectors and regions.[83] OECD's assessment of digital skills for private-sector competitiveness in Uzbekistan identifies the institutional gap clearly: there is no specialized agency responsible for providing structured digital literacy tools and adoption support to SMEs, existing incentives are concentrated in IT Park and IT- and BPO-sector priorities rather than being economy-wide, and regulatory barriers around personal data protection discourage even digitally capable firms from investing in digital transformation. In response, Uzbekistan should establish a dedicated one-stop-shop for SME digital transformation - analogous to the SME digital support frameworks that OECD countries have developed - that consolidates financial incentives, training programmes, certification schemes, and advisory services under a single institutional roof accessible to firms across all sectors and regions.

The GII 2025 data on Uzbekistan are encouraging in this respect: the country ranked 13th globally in graduates in science and engineering, which indicates a strengthening science and technology supply[84]. However, translating science graduates into productive digital innovators and technology entrepreneurs requires improved linkages between universities, the IT Park ecosystem, and the broader private sector. Kazakhstan's Astana Hub model offers a regional precedent for public-private technology commercialization, and Uzbekistan's IT Park has already begun developing analogous functions through its international offices and startup support programmes.

A practical step would be to develop a unified digital economy monitoring framework - with measurable annual indicators aligned to the GII, EGDI, and World Bank GovTech indices - that allows the government to track implementation progress systematically and adjust policy in response to data rather than only to political targets. Kazakhstan's use of international indices as benchmarks for policy evaluation offers a practical model for how Uzbekistan could embed evidence-based digital governance into its existing strategy architecture.

Potential effects on Uzbekistan

The most plausible and proximate expected outcome of successful private-sector digitalization is a productivity improvement at the firm level. When SMEs adopt digital tools - accounting software, e-commerce platforms, cloud-based management systems - they reduce administrative costs, improve access to markets, and generate economies of scale that are not otherwise achievable at small firm sizes. OECD evidence from both developing and transition economies confirms this linkage. For Uzbekistan, where SME productivity is currently constrained by limited digital adoption and weak digital skill levels, successful implementation of the SME-focused digital support framework recommended above could produce measurable productivity gains within a five-to-seven year horizon, which in turn would reduce the cost base of domestic firms and improve their competitiveness in both domestic and export markets[85].

The second expected outcome is improved public administration efficiency and reduced transaction costs for firms. As the e-government deepening

recommendations are implemented - particularly the back-end integration of government data systems and the development of machine-readable regulatory environments - the administrative burden on businesses should decline materially. This improvement would benefit all firms but disproportionately benefit SMEs and new entrants, for whom compliance costs represent a relatively heavier burden. Reduction in administrative transaction costs is one of the clearest and most policy-controllable channels through which digital investment affects growth.

The third expected outcome concerns trade integration and ICT services exports. Uzbekistan's ICT export trajectory - from USD 0.6 million in 2017 to USD 619.74 million in 2024 - demonstrates that the country already has a comparative advantage in digital services that is growing rapidly.[86] With deeper broadband infrastructure, stronger regulatory alignment with international standards, and an expanded IT talent pool, Uzbekistan has a realistic path toward the 2030 target of USD 5 billion in annual ICT exports. These export revenues would also contribute to current account improvement and strengthen macroeconomic stability, creating a more favourable environment for the broader digital and non-digital investment needed to sustain growth.

A fourth expected outcome is the attraction of higher-quality foreign direct investment. OECD's investment roadmap for Uzbekistan identifies digital transformation as a key factor in improving the country's investment profile, particularly for technology-intensive sectors. As Uzbekistan's regulatory framework becomes more predictable, data governance more internationally aligned, and digital infrastructure more reliable, the country becomes more attractive not only for IT services companies - which are already entering through IT Park - but also for manufacturing firms that use digital tools in supply chain management, logistics, and quality control. Kazakhstan's experience shows that this broader category of digitally-enabled FDI can support the diversification agenda more effectively than IT-sector FDI alone.

These outcomes are cumulative and interdependent: infrastructure quality enables productive firm use; firm-level digital adoption generates productivity gains; productivity gains improve competitiveness and exports; better competitiveness and governance attract investment; and investment supports deeper innovation. This sequence reflects the logic identified in the OECD and World Bank frameworks, and it explains why the short-run econometric evidence reviewed in Chapter II did not detect significant growth effects from digitalization indicators: Uzbekistan has advanced primarily through the first stages of this sequence - access and e-government - without yet reaching the deeper stages where aggregate growth effects become statistically detectable.

CONCLUSION

This graduation qualification work examined the impact of digitalization and the adoption of high technologies on economic growth in Central Asian countries, using Kazakhstan, Uzbekistan, and Kyrgyzstan as the main case studies. The research showed that digitalization is not only a technological process, but also an important factor of economic modernization, productivity improvement, innovation development, and long-term competitiveness.

The first part of the research studied the theoretical and methodological foundations of digitalization and economic growth. The analysis was based on major economic growth theories, including the Solow model, Romer's endogenous growth theory, and the Schumpeterian growth approach. These theories helped explain why technological progress, innovation, knowledge spillovers, and digital infrastructure can influence economic growth. In this study, digitalization was understood as a process that affects the economy through internet access, mobile connectivity, broadband infrastructure, productivity, innovation, trade integration, and institutional development.

The second part of the research focused on the comparative analysis of Kazakhstan, Uzbekistan, and Kyrgyzstan. The findings showed that all three countries made significant progress in digital development between 2004 and 2023. Internet use increased sharply in all selected countries. By 2023, internet penetration reached approximately 92.9% in Kazakhstan, 89% in Uzbekistan, and 88.5% in Kyrgyzstan. This shows that basic digital access has expanded strongly across the region. However, the quality and speed of digital transformation were not the same. Kazakhstan appeared as the early leader and benchmark country because of its more mature digital strategy, stronger institutional framework, and better digital government development. Uzbekistan showed rapid progress, especially after the acceleration of reforms in recent years, while Kyrgyzstan demonstrated more uneven digital development.

The research also systematized economic and digital indicators for the three countries over the period 2004-2023. The panel dataset included 3 countries, 20 years, and 60 observations. The main dependent variable was GDP growth, while the explanatory variables included internet users, mobile subscriptions, fixed broadband subscriptions, trade openness, and inflation. Descriptive analysis showed that GDP growth was unstable across the region. For example, the average GDP growth rate in the panel was around 3.6%, while the lowest value was recorded in Kyrgyzstan in 2020 at around -9%, mainly due to the COVID-19 shock. The highest growth was recorded in Kazakhstan in 2006 at around 9.8%. This confirms that economic growth in the selected countries was affected not only by digitalization, but also by external shocks and macroeconomic conditions.

The econometric analysis was conducted using pooled OLS, fixed effects, and random effects models. The main result was that inflation appeared as the only statistically significant variable across the main model specifications. In the pooled OLS model, the coefficient of inflation was approximately 0.1427, with a p-value of 0.0081. This means that inflation had a statistically significant positive relationship with GDP growth in the selected sample. The model's R-squared was 0.3142, meaning that about 31.4% of the variation in GDP growth was explained by the included variables.

The final chapter of the thesis focused on Uzbekistan and the adaptation of international experience. The analysis showed that Uzbekistan has made visible progress in digital transformation, especially through internet expansion, digital government, IT Park development, ICT service exports, and the Digital Uzbekistan 2030 strategy. However, the main challenge is to move from digital access to productive digital use. In other words, it is not enough to expand internet coverage; digital technologies must be actively used by businesses, public institutions, universities, and regional economies.

Based on the research findings, several practical recommendations were developed for Uzbekistan. First, the quality of digital infrastructure should be improved, especially fixed broadband and secure digital infrastructure. Second, digital adoption programs for small and medium-sized businesses should be expanded. Third, cooperation between universities, IT Park, startups, and the private sector should be strengthened. Fourth, regional digital inequality should be reduced so that the benefits of digitalization are not concentrated only in large cities. Fifth, human capital and digital skills should be developed, because digital technologies can create economic value only when people and firms are able to use them effectively.

Overall, the thesis concludes that digitalization is an important long-term factor of economic modernization in Central Asia, but its impact on economic growth is not automatic. The econometric results showed that digitalization indicators did not have a statistically significant direct short-term effect on GDP growth in the selected model. However, the theoretical, comparative, and policy analysis showed that digitalization remains essential for long-term development. Its real economic effect depends on complementary conditions such as infrastructure quality, digital skills, business adoption, innovation capacity, and institutional efficiency.

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APPENDICES

Appendix 1
Variable Definitions and Data Sources

Variable	Description	Measurement	Source
GDP_Growth	Economic growth indicator	Annual %	World Bank WDI
Internet_Users	Individuals using the internet	% of population	World Bank / ITU
Mobile_Subscriptions	Mobile cellular subscriptions	Per 100 people	World Bank / ITU
Fixed_Broadband	Fixed broadband subscriptions	Per 100 people	World Bank / ITU
Trade_GDP	Trade openness	% of GDP	World Bank WDI
Inflation	Inflation rate	Annual %	World Bank / IMF

Appendix 2
Conceptual framework linking digitalization and economic growth.
Source: Prepared by the author based on Solow (1957) and Romer (1990)

- <https://ru.scribd.com/document/685636688/R-Solow-Technical-Change-and-the-Aggregate-Production-Function>;
- https://papers.ssrn.com/sol3/papers.cfm?abstract_id=226703

Appendix 3
Pooled OLS output
Source: Prepared by the author

Appendix 4

Random Effect output
Source: Prepared by the author

Appendix 5
Effect output
Source: Prepared by the author

Appendix 6
ICT: Service export of Uzbekistan (2013-2024)
Source: <https://www.ceicdata.com/en/uzbekistan/telecommunication/uz-ict-service-exports-bop>

Source: <https://www.ceicdata.com/en/uzbekistan/telecommunication/uz-ict-service-exports-bop>

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